Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 28642

Part A: Each question is worth 3 points

1. The digits from 1 to 9 are written on 9 cards. Sara has 6, 5 and 4; Martha has 1, 2 and 9; Leo has 3, 7 and 8. Each of them uses some of +, –, ×, ÷ (and brackets) and each of their own cards exactly once.
Who *cannot* get 26 as the result?

A. Sara ***B.*** Martha ***C.*** Leo
D. All can get 26 ***E.*** Martha and Leo

2. To write all the room numbers in a hotel, starting from 1, exactly 129 digits are needed.
How many rooms are there?

A. 129 ***B.*** 80 ***C.*** 78 ***D.*** 69 ***E.*** 61

Part B: Each question is worth 4 points

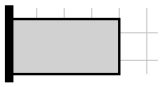
3. Tom builds a brick 7 cubes long and 4 cubes tall from red and blue cubes. The outside is completely red and the inside is completely blue. Tom used exactly 30 blue cubes.
How many cubes deep is the brick?

A. 6 ***B.*** 5 ***C.*** 4 ***D.*** 3 ***E.*** 2

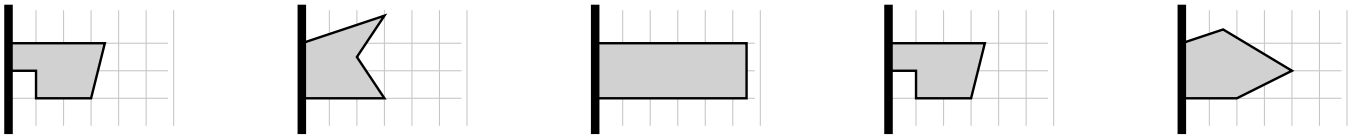
4. Sofia, Ivan, Aisha, Hana and Omar each play a different instrument: the harp, the drum, the flute, the trumpet and the piano. Sofia plays with sticks. Omar blows into the instrument. Hana presses keys. Omar doesn't play the trumpet. Ivan blows into the instrument.
Which statement is true?

A. Sofia plays the harp ***B.*** Aisha plays the flute ***C.*** Hana plays the trumpet
D. Omar plays the flute ***E.*** Omar plays the piano

5. From a window, Mia sees a rectangular flag on a pole. The wind turns and folds the flag, but the side sewn to the pole never moves. Mia draws the shadow of the flag on the wall at five different moments. Flat, the flag is 4 squares wide and 2 squares tall (see the picture). Some of the drawings are wrong.



the flag, flat



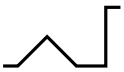
A B C D E

How many of the five drawings can be right?

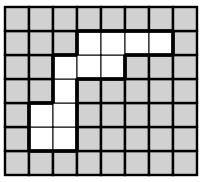
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

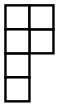
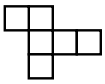
6. Far away we see the skyline of a castle.

Which of the pieces **can** belong to the skyline?

A. ***B.*** ***C.***  ***D.*** ***E.***

7. Two pieces that are exactly the same cover the empty area. Which piece is it? The pieces may be turned and flipped over.



A. ***B.*** ***C.***  ***D.***  ***E.***

8. Omar has three cards with different digits. Two of them are 4 and 3; the third is face down. Using each card exactly once, with +, –, ×, ÷ and brackets, Omar gets 29.
Which of these digits can be on the hidden card?

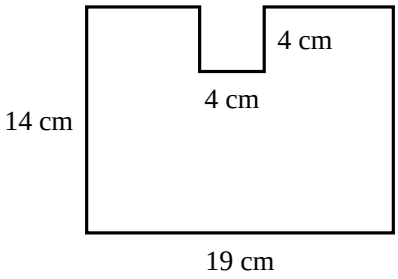
A. 9 ***B.*** 8 ***C.*** 7 ***D.*** 5 ***E.*** 2

9. Cleo has a rope 4 m long and cuts it into pieces that are each 40 cm long.
How many pieces does Cleo get?

A. 1 ***B.*** 8 ***C.*** 9 ***D.*** 10 ***E.*** 20

Part C: Each question is worth 5 points

10. A 19 cm by 14 cm rectangle of cardboard has a rectangular notch 4 cm wide and 4 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 82 cm ***B.*** 75 cm ***C.*** 74 cm ***D.*** 70 cm ***E.*** 66 cm

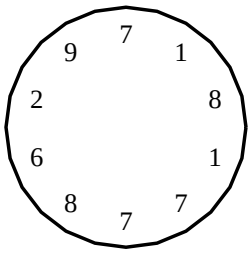
11. The odometer of a scooter shows 5689 km. Each digit is bigger than the digit before it.
After how many more kilometres will this happen again?

A. 90 ***B.*** 100 ***C.*** 109 ***D.*** 110 ***E.*** 1100

12. Exactly 375 digits are used to write all the house numbers in a street, starting from 1.
How many houses are there?

A. 161 ***B.*** 160 ***C.*** 125 ***D.*** 71 ***E.*** 62

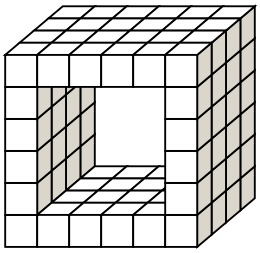
13. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



In how many different ways can he do it?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

14. Nina's tunnel is 4 cubes long. The hole goes straight through. Nina makes the tunnel longer, keeping the same front, until it is 8 cubes long.



How many cubes does Nina have to add?

A. 24 ***B.*** 60 ***C.*** 66 ***D.*** 71 ***E.*** 80

15. Dan's new rug is 1 m 8 cm long. Four of these lengths are the same as that.
Which one is *not* the same?

A. 10 dm 8 cm ***B.*** 1 m 80 mm ***C.*** 108 cm ***D.*** 108 mm ***E.*** 1080 mm

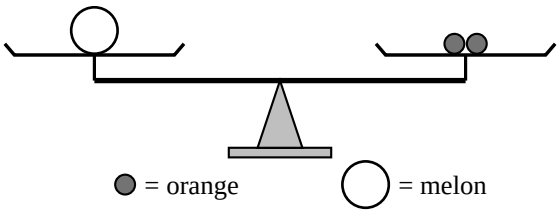
16. A bar code is made of 30 black and white bars taking turns. It starts with a black bar and ends with a **white** bar. Each black bar is wide or narrow. There are 11 more white bars than wide black bars.
How many narrow black bars are there?

A. 3 ***B.*** 4 ***C.*** 10 ***D.*** 11 ***E.*** 13

17. Five children, David, Grace, Anna, Clara and Felix, sit in a row of 5 chairs at the cinema. David and Grace are best friends and want to sit next to each other.
In how many different ways can the children sit?

A. 72 ***B.*** 50 ***C.*** 48 ***D.*** 8 ***E.*** 4

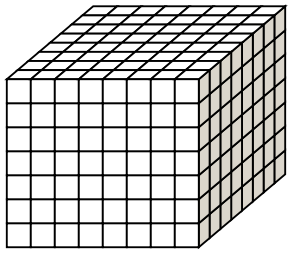
18. The balance shows that one melon weighs as much as 2 oranges. 1 melon and 2 oranges weigh 500 g altogether.



How much does one orange weigh?

A. 250 g ***B.*** 125 g ***C.*** 100 g ***D.*** 75 g ***E.*** 50 g

19. Nina builds a brick 8 cubes long, 8 cubes deep and 7 cubes tall. To make it strong, the red outside is **two** cubes thick everywhere (also at the bottom, top, front and back). Everything inside that is blue.



How many blue cubes are there?

A. 400 ***B.*** 100 ***C.*** 96 ***D.*** 48 ***E.*** 16

20. Ivan, Sofia, Tom, Julien and Nora were born in different months: July, August, December, October and January. Tom was born in autumn. Julien and Sofia both were born in a month that starts with J. Nora has a birthday in summer. Sofia has a birthday in summer.
Which statement is **not** true?

A. Ivan was born in December ***B.*** Tom was born in October ***C.*** Tom was born in December
D. Julien was born in January ***E.*** Nora was born in August

Answer Key

1.	**E**	**2.**	**D**	**3.**	**B**	**4.**	**D**	**5.**	**B**	**6.**	**B**
7.	**C**	**8.**	**B**	**9.**	**D**	**10.**	**C**	**11.**	**B**	**12.**	**A**
13.	**B**	**14.**	**E**	**15.**	**D**	**16.**	**D**	**17.**	**C**	**18.**	**B**
19.	**D**	**20.**	**C**								

Solutions

1. E (Martha and Leo) — Sara: 5 × 6 – 4 = 26. Martha: with 1, 2 and 9 nothing gives 26 (the closest are 20 = (1 + 9) × 2 and 27 = (1 + 2) × 9); Leo: with 3, 7 and 8 nothing gives 26 (the closest are 18 = 3 + 7 + 8 and 29 = 3 × 7 + 8). [2002 Q21 · who_cannot]

2. D (69) — 1 to 9 use 9 digits. The other 129 – 9 = 120 digits belong to two-digit numbers: 120 ÷ 2 = 60 numbers, from 10 to 69. So there are 69. [2003 Q16 · pages_from_digits]

3. B (5) — The blue inside is 5 long and 2 tall, so one layer of it has 5 × 2 = 10 cubes. 30 ÷ 10 = 3 layers of blue, plus the red front and back: 5. [2003 Q9 · how_deep]

4. D (Omar plays the flute) — The only possible matching: Sofia – drum; Ivan – trumpet; Aisha – harp; Hana – piano; Omar – flute. [2002 Q18 · which_true]

5. B (2) — A drawing is wrong if part of the pole edge is bare or if it reaches more than 4 squares from the pole. Wrong: A, C, D. So 2 can be right. [2002 Q15 · count_possible]

6. B — Only this piece is found in the line. The others don't fit — A: it steps down where the line steps up; C: it has a pointed roof where the line has a round dome; D: it has two small bumps where the line has a round dome; E: its flat part between the two features has the wrong length. [2002 Q7 · can_belong]

7. C — The empty area has 12 squares, so each piece has 6; every option has 6. Only two copies of piece C fit together into the shape of the hole. [2003 Q18 · two_copies]

8. B (8) — 4 × 8 – 3 = 29. The other digits do not work with 4 and 3. [2002 Q21 · hidden_card]

9. D (10) — 4 m = 400 cm. 400 ÷ 40 = 10. [2003 Q10 · pieces]

10. C (74 cm) — The bottom of the notch (4 cm) replaces the 4 cm taken out of the top edge, and the two sides of the notch add 2 × 4 cm. Perimeter = 2 × (19 + 14) + 2 × 4 = 74 cm. [2002 Q6 · notch]

11. B (100) — Keep as many digits on the left as possible. The next such number is 5789: 5789 – 5689 = 100. [2002 Q24 · going_up]

12. A (161) — 1-9 use 9 digits, 10-99 use 90 × 2 = 180, together 189. The other 375 – 189 = 186 digits are three-digit numbers: 186 ÷ 3 = 62, from 100 to 161. [2003 Q16 · three_digit_pages]

13. B (1) — Each half must have 28. The ways: 7+7+8+6 | 2+9+7+1+8+1. [2002 Q11 · count_ways]

14. E (80) — Each slice of the tunnel has 6 × 6 – 4 × 4 = 20 cubes. 4 new slices: 20 × 4 = 80. [2002 Q20 · longer_tunnel]

15. D (108 mm) — 1 m 8 cm = 108 cm = 1080 mm. But 108 mm = 108 mm. [2003 Q10 · not_equal]

16. D (11) — Starting black and ending white, the bars come in pairs: 15 black and 15 white. Wide: 15 – 11 = 4. Narrow: 15 – 4 = 11. [2003 Q17 · ends_differ]

17. C (48) — Think of David and Grace as one block: the block and the other 3 make 4 things to order, 4 × 3 × 2 × 1 = 24 ways. Inside the block David and Grace can swap, so 2 × 24 = 48. [2002 Q22 · side_by_side]

18. B (125 g) — Swap each melon for 2 oranges: 1 × 2 + 2 = 4 oranges weigh 500 g. One orange: 500 ÷ 4 = 125 g. [2002 Q9 · grams]

19. D (48) — Two layers come off at each end of every length, so each length drops by 4: 4 × 4 × 3 = 48. [2003 Q9 · thick_wall]

20. C (Tom was born in December) — The only possible matching: Ivan – December; Sofia – July; Tom – October; Julien – January; Nora – August. [2002 Q18 · which_not_true]